

Democratizing Urban Planning through Interactive Visualization: A Formative Study and System Design

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Abstract

Urban planning processes often struggle to achieve meaningful public participation due to technical complexity and communication barriers between professional planners and the general public. Through a formative study with urban planning professionals and community stakeholders, we identified key challenges in participatory urban planning and derived design requirements for an interactive system. Based on these insights, we developed an interactive visualization system that enables non-expert users to engage with urban planning concepts through real-time 3D visualization and intuitive parameter controls. Our preliminary user feedback from participants suggests that interactive visualization can help reduce technical barriers and enhance public understanding of planning decisions. This work contributes to participatory urban planning research by demonstrating how interactive systems can support democratic planning processes. Our ongoing work focuses on developing quantitative analysis tools, integrating professional planning features, and conducting comprehensive evaluation studies to further validate the system's effectiveness in real-world planning scenarios.

CCS Concepts

• **Applied computing** → **Computer-aided design**; • **Human-centered computing** → Participatory design.

Keywords

Urban planning, Generative design, Participatory design

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1 Introduction

Urban planning decisions significantly impact daily lives of citizens, yet meaningful public participation in planning processes remains a persistent challenge [1, 6, 8]. While the importance of citizen

engagement in urban planning is widely recognized, traditional participation methods often fail to bridge the gap between professional planners and the general public [2, 9]. This disconnect stems from three primary challenges: technical complexity of planning tools, communication barriers between experts and citizens, and the lack of intuitive platforms for real-time feedback.

In recent years, there have been increasing efforts to democratize urban planning through digital tools and participatory approaches. However, most existing solutions either focus on professional planning capabilities, making them inaccessible to the general public, or oversimplify planning concepts to the point where they lose practical utility for actual decision-making. This creates a critical need for systems that can effectively balance accessibility with planning rigor.

To address these challenges, we conducted a formative study with urban planners and community stakeholders to understand the barriers to effective public participation in planning processes. Our findings revealed that while citizens are eager to contribute to planning decisions affecting their communities, they often struggle to understand technical planning parameters and visualize the implications of different design choices. Similarly, planning professionals expressed frustration with the limitations of existing tools in facilitating meaningful dialogue with the public.

Based on these findings, an interactive visualization system was developed to empower non-expert users to engage with urban planning concepts (illustrated in Fig. 1). The following components are incorporated: 1. Real-time three-dimensional visualization of planning on a community scale. 2. Intuitive parameter controls that translate technical planning concepts into accessible terms. 3. Interactive feedback mechanisms that support dialogue between citizens and planning professionals.

Our research makes several contributions to the field of HCI research and urban planning practice. Firstly, we have developed a comprehensive understanding of the challenges in democratizing urban planning. This understanding is based on formative research with both professional planners and community stakeholders. Secondly, we have formulated design principles for interactive systems that bridge the gap between technical planning capabilities and public accessibility. Finally, we have presented initial evidence from preliminary user feedback suggesting how interactive visualization can support democratic planning processes while maintaining professional standards.

This paper presents our research journey from formative study through system design to preliminary user feedback. We begin by examining related work in participatory urban planning and interactive visualization systems. We then detail our formative study

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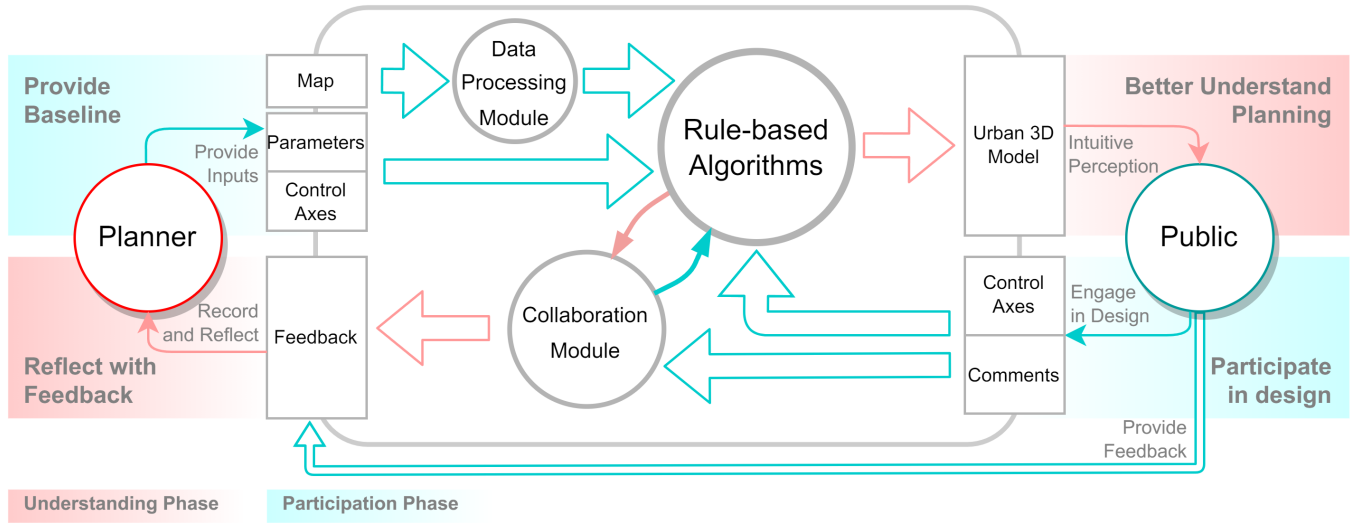


Figure 1: The workflow of our system between urban planning expert and public. This system helps the public and planners understand each other better and give feedback.

findings and how they informed our system design. Following a description of our system's key features, we present initial user feedback and discuss implications for future development. We conclude by outlining ongoing work to enhance the system's capabilities and evaluate its effectiveness in real-world planning scenarios.

Our research focuses specifically on community-scale planning decisions, where public engagement is most critical and the impact of decisions most directly affects residents' daily lives. This scope allows us to meaningfully explore how interactive visualization can support democratic planning processes while maintaining the technical rigor necessary for professional urban planning practice.

2 Related Work

Our work builds upon prior research in participatory urban planning and interactive visualization systems, focusing on participatory approaches and public engagement tools.

2.1 Participatory Approaches in Urban Planning

Participatory urban planning has its theoretical roots in Arnstein's ladder of citizen participation [1], which defines levels of public involvement. Collaborative processes [6] and communicative action [8] further emphasize democratic and inclusive planning. However, traditional methods often face technical and communication challenges [7].

Digital tools have expanded public participation possibilities. Pettit et al. [12] reviewed planning support systems for smart cities but noted their complexity for non-experts. Kontokosta [9] highlighted the accessibility gap in digital tools. Recent work has explored gamification and augmented reality (AR) to enhance engagement. Bhardwaj et al. [3] introduced a gamified AR tool for reconstructing urban spaces, while Sassmannshausen et al. [13] demonstrated AR's potential to motivate younger citizens in planning processes.

2.2 Interactive Visualization for Public Engagement

Interactive visualization is a powerful tool for participatory planning. Wissen et al. [15] demonstrated early integration of indicators into 3D visualizations, while recent advancements include augmented reality [11], real-time 3D systems [14], and tangible interfaces [11]. However, many systems overemphasize professional tools at the expense of public accessibility or fail to incorporate real-time feedback [5, 10].

New approaches address these limitations. Zhu et al. [16] explored situated visualization with socially acceptable interactions, while Dong et al. [4] proposed embodied interaction designs to connect users with urban spaces. These innovations demonstrate the potential of immersive technologies and user-centered design for enhancing public engagement.

Building on these advances, our work aims to balance accessibility and planning rigor by integrating real-time feedback mechanisms and quantitative tools into a participatory urban planning platform.

While participatory approaches aim for inclusive planning, and interactive visualization offers engagement tools, a gap remains in integrating these effectively. Existing planning support systems, as noted by Pettit et al. [12], often lack accessibility for non-experts. Our work directly addresses this gap by synthesizing principles from collaborative planning and communicative action with the capabilities of interactive 3D visualization. Our system is designed to empower citizens with accessible tools that still allow for meaningful interaction with complex planning parameters. It is distinct from gamified approaches or AR applications in its focus on direct manipulation of planning parameters and real-time feedback within a collaborative 3D environment.

3 Formative Study

To ensure that our system addresses the needs of both urban planners and the general public, we conducted a formative study to understand the challenges in participatory urban planning. This study was conducted in the greater bay area in China, a metropolitan area with a diverse population and a history of active community engagement in planning processes. Participants were recruited through a combination of methods, including online and onsite advertisements targeting local neighborhood groups, and direct outreach to planning firms and professional organizations. Our study included interviews with 20 urban residents (R1-R20, ages 25-65, have diverse educational backgrounds) and 8 planning professionals, including urban planners (P1-P3), architects (P4-P6), and community engagement specialists (P7-P8) from both public and private sectors. The purpose of the study was to gain insights into the current challenges faced by urban planners and residents regarding the urban planning process and outcomes, as well as to identify desired functionalities for an interactive urban planning tool. Our study consisted of three phases: semi-structured interviews with urban residents and planning professionals, analysis of existing tool usage, and synthesis of key design requirements.

3.1 Interviews with Urban Residents and Planners

Our interviews were structured to explore critical areas relevant to both urban residents and professional urban planners. For urban residents, we focused on their perceptions of the professional barriers to urban planning for the general public, the level of engagement they felt they had in the urban planning process, and their likely willingness to participate when accessible tools for participation are available. For urban planners, the focus was on the tools and methodologies they used in their day-to-day work and the perceived barriers to more effective public participation in urban planning.

Our findings from resident interviews revealed significant challenges in community participation. While most participants (17/20) expressed strong interest in community planning, they consistently reported difficulties with technical complexity. As R3 noted, "I find the planning documents and terminology overwhelming." Many residents (15/20) struggled with visualization aspects, with R7 explaining, "It's hard to imagine how proposed changes will actually look." The lack of effective feedback channels was another common concern, as expressed by R11: "I never know if my input actually makes a difference."

Planning professionals highlighted different yet complementary challenges. Urban planners (P1-P3) emphasized the difficulty in translating technical concepts for public understanding without oversimplification. Architects (P4-P6) noted challenges in effectively communicating spatial design concepts to non-experts. Community engagement specialists (P7-P8) stressed the need for better integration of qualitative feedback with quantitative analysis in planning processes.

3.2 Analysis of Existing Tools

We analyzed current planning tools through observation sessions with both residents and professionals. Particular attention was

paid to widely-used platforms such as ArcGIS for 2D planning visualization and SketchUp for 3D modeling. Our analysis of ArcGIS revealed that while it offers powerful spatial analysis capabilities, its complex interface and technical requirements create significant barriers for public users. Most residents (19/20) found the layering system and technical terminology confusing, limiting their ability to understand basic spatial relationships.

SketchUp, presented challenges in collaborative settings. While professionals could effectively create 3D models, the software's limited real-time collaboration features hindered immediate feedback collection. Online feedback platforms, primarily consisting of surveys and voting systems, showed limitations in capturing nuanced spatial feedback and failed to provide meaningful visualization of proposed changes.

The key limitations across these tools included the disconnect between visualization and feedback mechanisms, the inability to support real-time group interactions, and the challenge of balancing professional planning capabilities with public accessibility. These findings highlighted the need for an integrated approach that combines visualization, interaction, and feedback collection in a more accessible format.

3.3 Design Requirements

Based on our findings, we identified five key requirements for our system: (1) accessible visualization through intuitive 3D interfaces, (2) interactive feedback mechanisms that connect individual inputs with collective discussions, (3) real-time collaboration support for group engagement, (4) simplified access to professional planning capabilities, and (5) integration of quantitative analysis tools to support evidence based decision making.

These requirements emerged from our formative study findings and guided our subsequent system design decisions. They reflect both the immediate needs expressed by the participants in our study and the broader goals to improve the participatory planning processes identified in previous research.

4 System Design

Our system design directly addresses the challenges identified in the formative study, specifically the need to bridge the gap between technical complexity and public accessibility. Each module was designed with a focus on translating complex planning processes into intuitive interactions, while maintaining the fidelity required for professional planning tasks. The following sections detail the four primary modules and their mapping to the design requirements.

4.1 Data Processing Module

The data processing module handles urban data from OpenStreetMap to create a foundation for planning generation. For road networks, the system processes and simplifies the complex road hierarchy into four categories: highways, main roads, secondary roads, and branch roads. The module extracts essential building attributes including boundaries, functions, and heights from the raw data. Users can define planning areas by drawing polygons on the map, with the system automatically processing the enclosed urban elements for subsequent planning operations.

4.2 Planning Generation Module

In order to simplify input for non-expert users, while being able to reflect their main ideas, the planning generation module employs rule-based algorithms to transform user basic input parameters into concrete urban planning schemes. The road network generation begins with user-defined control axes, which serve as the primary spatial organization elements. Through recursive subdivision algorithms, the system generates a hierarchical road network that responds to local context and planning requirements. The module then divides the area into blocks of varying scales based on the road network structure. Building layout generation considers multiple planning indicators, including functional ratios and floor area ratios, to create balanced and realistic urban environments.

4.3 3D Visualization Module

The visualization module transforms generated planning schemes into interactive three-dimensional scenes, making abstract planning concepts tangible for non-expert users. Road visualization incorporates varying widths according to road hierarchy and includes pedestrian spaces along block boundaries. Building visualization generates models with different typologies and uses color coding to represent functional categories. The visualization interface supports fundamental interaction operations including zoom, pan, and rotation, allowing users to examine planning proposals from multiple perspectives. The module emphasizes visual clarity and intuitive interaction to enhance accessibility for public users while maintaining sufficient detail for professional planning purposes.

4.4 Collaboration Module

To support participatory planning processes, we integrated a collaboration module that enables multi-stakeholder engagement. The module facilitates real-time collaboration through synchronized viewing and editing of planning proposals. Users can add annotations and comments directly within the 3D environment, creating spatial references for discussion points. The system provides side-by-side comparison views for different planning scenarios, enabling effective evaluation of alternatives. An automated feedback analysis system aggregates user input and generates visual reports, helping planners understand public preferences and concerns. This module particularly emphasizes bidirectional communication between professional planners and the public, supporting democratic planning processes.

4.5 User Interface

The system interface design (illustrated in Fig. 2) prioritizes clarity and accessibility while maintaining professional functionality. The main interface consists of three primary regions: a parameter control panel providing access to planning parameters, a 3D scene view for spatial visualization, and a collaboration panel for feedback and discussion. The interface responds dynamically to user interactions, providing immediate visual feedback for parameter adjustments and planning modifications. This design approach supports both detailed professional planning work and accessible public participation.

Through this modular design, our system addresses both professional planning requirements and public accessibility needs. The

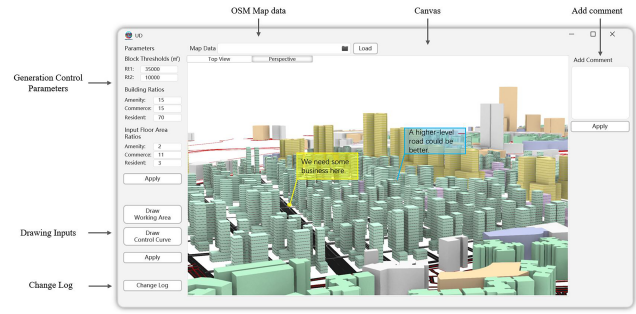


Figure 2: The graphical interface of our system.

tight integration between modules provides technical support for democratized urban planning processes, enabling meaningful participation while maintaining planning rigor. The system's architecture reflects our commitment to balancing professional capabilities with public engagement, as identified in our formative study.

5 Preliminary User Feedback

To assess the initial effectiveness of our system in supporting participatory urban planning, we conducted preliminary user feedback sessions with 12 participants (8 residents without professional planning background and 4 urban planning professionals). Participants were recruited in the same way as formative study. The sessions focused on evaluating system usability and its potential for facilitating public participation in urban planning processes.

5.1 Study Setup

Participants were given a brief introduction to the system followed by hands-on experience with a sample planning scenario involving a local neighborhood redevelopment project. The session included three main tasks: (1) exploring and understanding an existing planning proposal through the 3D visualization interface, (2) making modifications to the proposal using the parameter controls, and (3) providing feedback through the collaboration module. Each session lasted approximately 45 minutes, followed by a semi-structured interview.

5.2 Initial Findings

Our preliminary findings suggest that the system shows promise in reducing barriers to public participation in urban planning. Non-expert participants (6/8) reported that the 3D visualization helped them better understand planning proposals compared to traditional 2D representations. As one participant noted, "Being able to see the buildings and streets from different angles made it much easier to grasp what the changes would mean for the neighborhood."

The interactive parameter controls received mixed feedback. While most participants appreciated the immediate visual feedback, some (3/8) non-expert users found certain planning parameters challenging to understand. This suggests a need for additional contextual information and simplified parameter descriptions in future iterations.

Professional users particularly valued the collaboration features, with all four highlighting the potential for gathering more structured public feedback. One planner remarked, "The ability to see public comments directly in the 3D context helps us better understand community concerns about specific aspects of the proposal."

5.3 Limitations and Future Work

These preliminary findings should be interpreted with caution given the limited sample size and controlled setting. Additionally, the brief interaction time may not fully reflect real-world usage scenarios. Our ongoing work focuses on addressing the identified usability challenges and conducting more comprehensive evaluation studies with larger user groups in actual planning contexts.

Key areas for improvement include enhancing parameter explanation for non-expert users, developing more intuitive interaction methods for plan modification, and expanding the quantitative analysis tools to support evidence-based decision making. We are also exploring integration with existing professional planning workflows to better support real-world implementation. Crucially, future work will include a comparative user experience (UX) evaluation, pitting our system against existing tools to assess relative usability and effectiveness in facilitating participatory planning. This evaluation will involve both expert and non-expert users, with specific tasks designed to measure the efficiency, accuracy, and satisfaction of each group when using different tools.

6 Discussion and Future Work

Our preliminary work demonstrates the potential of interactive visualization systems to support democratic urban planning processes while highlighting several key areas for future development and research.

6.1 Implications for Participatory Planning

The initial feedback suggests that interactive visualization can effectively bridge the technical gap between professional planners and the public. The system's ability to provide immediate visual feedback appears to enhance public understanding of planning proposals, addressing a key challenge identified in our formative study. However, the mixed responses to parameter controls indicate that further work is needed to balance technical precision with public accessibility.

6.2 Current Limitations

Several limitations in our current implementation warrant discussion. First, while the system supports basic planning operations, it lacks some advanced features required for comprehensive professional planning work. Second, the current evaluation focuses primarily on initial user reactions rather than long-term engagement effectiveness. Third, the system's computational requirements may limit its accessibility in some community settings.

6.3 Future Development

Our ongoing work focuses on three primary areas:

Quantitative Analysis Tools. We are developing additional analytical capabilities to help users understand the implications of

planning decisions through metrics such as walkability scores, solar access, and traffic flow predictions.

Professional Planning Integration. Future iterations will incorporate more advanced planning features while maintaining interface accessibility, including zoning regulation compliance checks and environmental impact assessments.

Comprehensive Evaluation. We plan to conduct longitudinal studies in actual community planning processes to evaluate the system's effectiveness in supporting democratic planning decisions. This will include quantitative measures of planning outcomes (e.g., comparing plans generated with and without the system) as well as qualitative assessments of participant satisfaction and perceived influence on the planning process. We will also compare the effectiveness of planning and the user experience between experts and non-experts using our system, and between our system and traditionally used planning tools.

7 Conclusion

This paper presents initial work toward democratizing urban planning through interactive visualization. Our formative study identified key challenges in participatory planning, informing the development of a system that combines accessible visualization with professional planning capabilities. Preliminary user feedback suggests that our approach shows promise in supporting democratic planning processes, while also highlighting areas for improvement.

The main contributions of this work include: (1) empirical insights into the challenges of democratizing urban planning, (2) a system design that balances public accessibility with professional planning requirements, and (3) initial evidence of the potential for interactive visualization to support participatory planning processes.

As cities face increasingly complex planning challenges, tools that support democratic decision-making become increasingly important. Our ongoing work aims to further develop and validate approaches that can make urban planning more accessible and participatory while maintaining professional rigor.

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